

THE AI LEARNING ROADMAP

A serial, step-by-step path — from “What is Intelligence?” to modern AI agents

By Shiwansh Rai

PHASE 1 THE FOUNDATION — WHAT IS INTELLIGENCE?

1. What Is Intelligence? — Not knowing facts — it's learning from experience and using that learning to make a decision (the dog-and-hot-stove example).

PHASE 2 MACHINE LEARNING & AGENTS

- 2. Machine Learning** — Teaching a computer by showing it examples instead of writing exact rules.
- 3. The Three Types of ML** — Supervised (learning with an answer key), Unsupervised (finding patterns with no answer key), Reinforcement (learning by trial and reward — the self-driving car example).
- 4. AI Agents** — Sensors (how it senses the world) + Actuators (how it acts on the world), looping through Sense → Think → Act.

PHASE 3 THE BUILDING BLOCKS — PERCEPTRONS & NEURAL NETWORKS

- 5. The Perceptron** — The first artificial neuron — a mathematical copy of a brain cell; the true foundation of AI, built to mimic the human brain.
- 6. Activation Functions** — The “decision gate” after the weighted sum (step, sigmoid, ReLU) that decides how strongly a perceptron fires.
- 7. Neural Networks** — Thousands of perceptrons stacked in layers (input → hidden → output), each layer refining the answer.
- 8. Weights & Training** — The 5-variable decision analogy — training means adjusting millions of weights, again and again, until answers match reality.

PHASE 4 NATURAL LANGUAGE PROCESSING → TRANSFORMERS → GPT

- 9. NLP Basics** — Tokenization, POS tagging, stemming/lemmatization (rose → roses), named entity recognition, sentiment analysis.
- 10. LSTM** — A neural network with a built-in memory, remembering important words from earlier in a sentence.
- 11. “Attention Is All You Need” (2017)** — The landmark paper that let models look at an entire sentence at once — the foundation of Transformers.
- 12. Multi-Headed Attention** — Running the attention trick several times in parallel, each “head” focusing on a different relationship between words.
- 13. GPT** — Generative Pre-trained Transformers — trained to predict the next word, which lets them write, answer, and converse.
- 14. Modern Neural Network Types** — CNN → images, RNN/LSTM → sequences, Transformer → language, GAN → generating new content.

PHASE 5 COMPUTER VISION

- 15. Computer Vision Basics** — CNNs scan small patches of an image, building edges → shapes → objects.
- 16. Vision Models & Tools** — ResNet50 (accuracy-focused), MobileNet (light, for phones/edge devices), OpenCV (the classic image-processing toolbox) — and when to use each.

PHASE 6 THE MODERN AI STACK

- 17. Agentic AI & AI Agents** — AI that plans and carries out multi-step actions on its own, not just answering once (Anthropic's Claude is one example of an AI assistant built this way).
- 18. Open Source vs Closed Source Models** — Open source: the weights are public, downloadable, and editable (e.g. Llama, Mistral). Closed source: usable only via an app/API (e.g. GPT, Claude).
- 19. Embeddings** — Turning text into a list of numbers so a computer can measure how close two meanings are.
- 20. RAG (Retrieval-Augmented Generation)** — Fetching real, relevant documents first, then handing them to the model — an “open-book exam” instead of guessing from memory.
- 21. The RAG Pipeline** — Chunk documents → turn into embeddings → store in a vector database → retrieve closest matches → generate a grounded answer.
- 22. Fine-Tuning vs Prompting vs RAG** — Three ways to customize a model — prompting (fastest), RAG (fresh facts), fine-tuning (a genuinely new skill or style).
- 23. AI Harness** — The software scaffolding — tools, memory, a repeating loop — that turns a raw model into an agent that can actually take action.
- 24. Multimodal AI** — Models that see, hear, and act together — e.g. Google's Project Astra, a camera-and-voice AI assistant.
- 25. Offline / Local Models** — Running a (quantized, shrunk-down) model on your own device — full privacy, no internet needed, no per-question cost.

PHASE 7 PUTTING IT ALL TOGETHER

- 26. How a Model Is Actually Trained** — Random weights → guess → measure how wrong → nudge every weight slightly → repeat billions of times. That loop is the whole of AI.

— *Shiwansh Rai*